

CHAEYEON KIM

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EDUCATION

University of Wisconsin-Madison

Ph.D. in Educational Psychology (Advisor: Dr. Mitchell Nathan)

Madison, United States

Sep. 2023 ~

Chonnam National University

M.A. in Educational Technology (Advisor: Dr. Jeeheon Ryu)

Gwangju, South Korea

Sep. 2019 – Feb. 2022

- GPA of 4.25/4.5
- Master's Thesis: *The Difference in Emotions and Engagement between Pre-service and In-service Teachers and their Genders with the Scenario Cycle of a Teaching Simulation for Classroom Management.*
- Visiting Scholar, University of South Florida, Jan. 2020 – Mar. 2020.
- Academic Excellence Scholarship for all semesters

Chonnam National University

B.Ed. in Education (Minor: Psychology)

Gwangju, South Korea

Mar. 2016 – Aug. 2019

- GPA of 4.04/4.5 and ranked 2/16 in an integrated B.A./M.A. program
- Academic Excellence Scholarship for all semesters

JOURNAL PUBLICATIONS

International Publication

- Moon, H., Yoon, H., Park, S., **Kim, C.**, Jeong, M., Lim, S., Ryu, J. & Heo, H. (2021). Usefulness of virtual reality-based training to diagnose strabismus. *Scientific Reports*, 11(1), 1-7. <https://doi.org/10.1038/s41598-021-85265-8> [PDF] (SCI-indexed)

Domestic Publications (Korean)

- **Kim, C.** & Ryu, J. (2022). The differences in emotional experiences of pre-service teachers and in-service teachers in a teaching simulation for classroom management. *The Korean Journal of Thinking Development*, 18(1), 27-46. <https://doi.org/10.51636/JOTD.2022.04.18.1.27> [PDF]
- **Kim, C.**, Kim, K., & Ryu, J. (2020). The effects of augmented reality based interaction and scenarios on virtual presence in teaching simulation. *Teacher Education Research*, 59(4), 615-630. <https://doi.org/10.15812/ter.59.4.202012.615> [PDF]
- **Kim, C.**, Kim, K., & Ryu, J. (2020). The effect of experience level on user perception in the VR based simulation for communication training with virtual patient. *The Journal of Educational Information and Media*, 26(3), 455-475. <https://doi.org/10.15833/KAFEIAM.26.3.455> [PDF]
- Yang, E., **Kim, C.**, & Ryu, J. (2019). The ethical guidelines on the affective model for artificial intelligence based autonomous avatar in teaching simulation. *The Journal of Korea Society of Internet Ethics*, 3(2), 24-34. <https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE10753966> [PDF] (3rd Prize)

BOOK CHAPTER

- Park, S., Ryu, J., Kim, K., & **Kim, C.** (2021). Mobile simulation for effective classroom management skill training for preservice teachers. In B. Hokanson, M. Exter, A. Grinciewicz, M. Schmidt & A.A. Tawfik (Eds.). *Learning: Design, Engagement and Definition* (pp. 223-238). Springer, Cham. https://doi.org/10.1007/978-3-030-85078-4_18 [PDF]

AWARDS

Michael B. Salwen Scholarship, 2026 KAERA Annual Meeting

Mar. 2026

Educational Psychology Student Travel Awards, The Department of Educational Psychology

Feb. 2026

School of Education Travel Awards , The Office of Research & Scholarship	<i>Jan. 2026</i>
Best Doctoral Colloquium Paper Award , 11th International Conference of the Immersive Learning Research Network (iLRN2025)	<i>Jun. 2025</i>
Student Research Grants Competition Travel Award , UW-Madison Graduate School	<i>Jun. 2025</i>
International Summer Research Award , The School of Education Global Engagement Office	<i>Apr. 2025</i>
Educational Psychology Student Travel Awards , The Department of Educational Psychology	<i>Feb. 2025</i>
Korean Government Fellowship for Study Overseas , Jeollanam-do Institute for Lifelong Education	<i>Apr. 2023</i>
Two Year University Graduate School Fellowship , the Office of the Vice Chancellor for Research and Graduate Education at the University of Wisconsin-Madison	<i>Feb. 2023</i>
Best Paper Award , 2021 Spring Conference of the KEST	<i>May. 2021</i>
Best Graduate Student Award , BK21 Flourishing Education Research Center, Chonnam National University	<i>Mar. 2021</i>
Young Scholar Award , International Conference for Media in Education	<i>Aug. 2020</i>
3rd Prize , 2019 Korea Society of Internet Ethics Research Paper Contest	<i>Dec. 2019</i>
Young Scholar Award , International Conference for Media in Education	<i>Jul. 2019</i>
1st Prize at Media Festival , 2019 Spring Conference of the KAEIM	<i>Jun. 2019</i>
Young Scholar Award , International Conference for Media Education	<i>Aug. 2018</i>
Encouragement Award , President Honor Students' Team Project, Chonnam National University	<i>Feb. 2018</i>
1st Prize at Media Festival , 2017 Autumn Conference of the KAEIM	<i>Oct. 2017</i>
2nd Prize at Media Festival , 2017 Spring Conference of the KAEIM	<i>Mar. 2017</i>

WORK AND RESEARCH EXPERIENCE

Collaborator, Multimodal STEM+AI Classrooms: A Sociocultural Inquiry into Participatory Design for Collaborative Learning *Mar. 2026 – Mar. 2027*

- PI: Hakeoung Hannah Lee, University of Virginia
- UVA School of Education and Human Development IDEAs Awards.

Teaching Assistant, University of Wisconsin-Madison *Jun. 2024 – May 2026*

- Assisted in large classes, including one online and two in-person courses, serving over 100 students.
- Delivered a 75-minute lecture on Embodied Cognition.

Instructor, Summer Transitional Enrichment Program (STEP) at UW-Madison *June 15 – 20, 2025*

- Taught *Mysteries of the Mind*, a one-week accelerated psychology course for advanced middle school students (grades 6–8), incorporating interactive lessons on cognition, brain function, and scientific thinking.

Instructor, Precollege Academic Campus Experience (PACE) at UW-Madison *July 7 – 11, 2025*

- Led *Neuropsychology*, an enrichment course introducing middle school students (grades 5–8) to brain structure, mental processes, and applied neuroscience through hands-on activities and discussions.

Reviewer for DPI's 2025 Gifted/Talented Grant Competition, Wisconsin *May 19 – 20, 2025*

- Independently reviewed and scored five different grant applications.
- Participated in consensus meetings.

Research Assistant, Chonnam National University (Advisor: Dr. Jeeheon Ryu) *Mar. 2017 – Feb. 2022*

- Participated in 10 research projects for the central government and institutions of learning.
- Designed and developed VR/AR/XR-based simulations through Unity 3D, DAZ 3D, iClone, and Vuforia.
- Collected real-time user log data (demographic data, user performance) through Firebase.
- Analyzed learner affective/cognitive states via self-reported survey, eye-tracking analysis, facial expression analysis.
- Assessed and modified simulations by evaluating learners' experiences through a mixed methods approach.
- Submitted resulting papers in six peer-reviewed journals and 35 conferences.

Teaching Assistant, Chonnam National University *Sep. 2019 – Feb. 2022*

- Assisted large online classes of over 150 students about *Multimedia Learning*.
- Provided one-on-one technical support to faculty and students in Metaverse (Virbela) online classes.

SELECTED CONFERENCE PRESENTATIONS

Recent Research

- Investigated how learners collaboratively ground and transfer mathematical ideas using gestures in extended reality-based geometry learning simulations (*The Hidden Village-Online, GeoGebra MR*).
 - Examined and compared teachers' interactions and discourse with students in AR-based simulations for geometric education.
 - Developed a coding scheme to analyze affordances and constraints of AR in geometric learning.
 - Reviewed and explored effective design features of extended reality-based learning simulations.
- **Kim, C.** (in press). Generative AI as a Cognitive Artifact: Rethinking Distributed Intelligence in University Writing. ICOLSEI 2026, Seoul, South Korea.
 - Lee, H. H., Kwon, H. G., Kim, J., Sung, J., Choi, J., Yang, H., **Kim, C.**, & Lujan, M. (under review; submitted March 2026). Influence and negotiation in co-designing an AI-enhanced MMLA platform for secondary STEM classrooms. In Companion Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED '26), Seoul, South Korea.
 - Doy, K., **Kim, C.**, & Nathan, M. J. (2026). What Makes Actions Matter in Virtual Reality? Examining the Effects of Conceptual Congruence and Sensorimotor Engagement on Geometric Reasoning. iLRN 2026, Athens, Greece.
 - **Kim, C.**, Doy, K., & Nathan, M. J. (2026). A Comparison of Physical and Mixed Reality Manipulatives on Geometric Reasoning. KAERA 2026, Los Angeles, CA.
 - **Kim, C.**, Swart, M. I., & Nathan, M. J. (2026). Embodied Geometric Reasoning in Augmented Reality: A Video Coding Framework for Cognitive Functions. ISLS 2026, Irvine, CA.
 - **Kim, C.**, Swart, M. I., & Nathan, M. J. (2026). Comparing Inductive and Deductive Geometric Reasoning in Augmented Reality. TEI 2026, Chicago, IL. <https://doi.org/10.1145/3731459.3779036>
 - **Kim, C.**, Lim, J., & Moon, J. (2026). A Systematic Review of Learning Analytics-Supported Feedback Practices in K-12 [Round Table]. 2026 AERA, Los Angeles, CA.
 - Yang, W., **Kim, C.**, & Ponto, K. (2025). Should we break the illusion? A study on non-diegetic audio in immersive informal learning. VRST 2025, Montreal, Canada. <https://doi.org/10.1145/3756884.3765969>
 - Nathan, M. J., Swart, M. I., Luu, R., Grondin, M. M., **Kim, C.**, Beier, J. P., Xia, F., Kemper, R. D. (2025). Direction, Magnitude & Torque: Evidence for embodied scaffolds' impact on understanding and transfer of learning. *Proceedings of Annual Conference of the Cognitive Science Society*, San Francisco, CA.
 - **Kim, C.**, Swart, M. I., Beier, J. P., Xia, F., Grondin, M. M., Kim, D., & Nathan, M. J. (2025). Doctoral Colloquium—A Comparison of Teachers and Students in Categorizing and Representing Geometric Conjectures using AR Simulations. *Immersive Learning Research - Academic*, 1(1), 168-175. <https://doi.org/10.56198/3ad2ea07> (**Best Doctoral Colloquium Paper Award**)
 - **Kim, C.**, Swart, M. I., Xia, F., Kim, D., & Nathan, M. J. (2025). Exploring Teachers' Perspectives on the Affordances and Constraints of Augmented Reality in Geometry Education. *Immersive Learning Research - Academic*, 1(1), 124-133. <https://doi.org/10.56198/gfpc9h96>
 - **Kim, C.**, Xia, F., Beier, J. P., Swart, M. I., Schenck, K. E., & Nathan, M. J. (2025). How Students Ground Mathematics Through Distributed Gestures: Designing Computer-Supported Collaborative Embodied Learning. In Oshima, J., Chen, B., Vogel, F., & Järvelä, J. (Eds.), *Proceedings of the 18th International Conference on Computer-Supported Collaborative Learning - CSCL 2025* (pp. 355-359). International Society of the Learning Sciences. <https://doi.org/10.22318/csc12025.241378> (**Nominated for the best student paper**)
 - Xia, F., **Kim, C.**, Beier, J. P., Swart, M. I., Schenck, K. E., & Nathan, M. J. (2025). Embodied Transfer of Mathematical Ideas Through Gestural Replay. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), *Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025* (pp. 917-925). International Society of the Learning Sciences. <https://doi.org/10.22318/icls2025.890730>
 - **Kim, C.**, Lim, J., Kang, M., & Moon, J. (2025). *A Systematic Literature Review on Design Features of Scenario-Based Learning in Virtual Reality* [Poster Presentation]. 2025 AERA, Denver, CO. [\[Link\]](#)

Multimodal Learning Analytics in Human-Computer Interaction

- Measured learner eye movement via iMotions eye tracking module (screen-based, glasses-based), FOVE headset.
 - Examined learner facial expression through iMotions facial expression analysis module.
 - Processed and linked physiological data to cognitive/affective states.
- **Kim, C.**, & Ryu, J. (October, 2022). *Comparison of the emotional responses of pre-service and in-service teachers in a teaching simulation* [Concurrent]. 2022 AECT. [\[PDF\]](#)

- **Kim, C., & Ryu, J.** (November, 2021). *The effect of teaching position on engagement, attention, and valence according to the scenario cycle of teaching simulation* [Concurrent]. 2021 Autumn Conference of KSET & KAEIM.
- **Kim, C., & Ryu, J.** (October, 2021). *Application of the facial expression recognition system for the emotional analysis of pre-service teachers in teaching simulations* [Poster]. 2021 SITE. [\[Link\]](#) [\[PDF\]](#)
- **Kim, C., & Ryu, J.** (August, 2021). *Exploring the emotional experience of in-service teachers through facial expression analysis in a teaching simulation* [Round Table]. 2021 ICoME.
- **Kim, C., & Ryu, J.** (May, 2021). *A study on measuring teacher's emotional experience through facial expression analysis in teaching simulation* [Poster]. Spring Conference of KSET.
- **Ryu, J., Kim, K., & Kim, C.** (March, 2019). *The effect on the scenario level on the eye-fixation time on the virtual problem student avatar in virtual classroom simulation* [Concurrent]. 2019 SITE. [\[Link\]](#) [\[PDF\]](#)
- **Ryu, J., Kim, K., & Kim, C.** (February, 2019). *Where do you look at a student avatar in the virtual reality based conversation?* [Poster]. 2019 ICoME.

Immersive Simulations for Improving Communication Skills of Teachers (SimTEACHER) [\[Link\]](#)

- Developed VR/AR simulations through Unity 3D, iClone, DAZ 3D, and Vuforia.
- Categorized 4 types of problematic student behaviors and developed 15 scenarios with in-service teachers.
- Assessed and modified simulations by evaluating learners' experiences.
- **Yang, E., Kim, C., & Ryu, J.** (June, 2021). *Work-in-progress-effects of interactive conversation on in-service teacher experience in classroom simulation for teacher training* [Poster]. 2021 iLRN. [\[PDF\]](#)
- **Park, S., Ryu, J., Kim, C., & Kim, K.** (November, 2020). *Design and validation of culturally-contextualized classroom management scenarios for virtual teaching simulation* [Concurrent]. 2020 AECT. [\[PDF\]](#)
- **Kim, K., Kim, C., Park, S., Ryu, J., & Lee, S.** (November, 2020). *Simteacher mobile: Design and development of immersive classroom management simulation for pre-service teacher training* [Media Showcase]. 2020 AECT. [\[PDF\]](#)
- **Kim, C., & Tak, Y.** (November, 2020). *Augmented reality for mobile simulation* [Round Table]. 2020 Asian Student Seminar Round Table (ASSERT). **(Young Scholar Award)**
- **Kim, C., Kim, K., & Ryu, J.** (October, 2020). *The effect of scenarios and interaction types on virtual presence in augmented reality-based classroom simulation* [Concurrent]. 2020 Autumn Conference of KSET & KAEIM.
- **Kim, K., Kim, C., Kim, D., Ryu, J.** (August, 2020). *Simteacher mobile: Design and development of augmented reality simulation for preservice teacher classroom management skill training* [Round Table]. 2020 ICoME. [\[PDF\]](#)
- **Park, S., Ryu, J., Kim, K., & Kim, C.** (July, 2020). Design and development of mobile simulation prototype for preservice teacher's classroom management skill training. In B. Hokanson, M. Exter, A. Grincewicz, M. Schmidt & A.A. Tawfik (Eds.). *Learning: Design, Engagement and Definition* [Symposium]. 2020 AECT SRS. [\[PDF\]](#)
- **Park, S., Ryu, J., Kim, K., & Kim, C.** (June, 2020). *Simteacher mobile: Augmented reality-based mobile simulation for preservice teacher classroom management skill training* [Media Showcase]. 2020 iLRN. [\[Link\]](#) [\[Video\]](#)
- **Yang, E., Kim, K., Kim, C., Jung, M., & Ryu, J.** (November, 2019). *Development of an artificial intelligence interactive avatar model that causes emotional conflict in class simulation* [Poster]. 2019 Autumn Conference of KSET & KAEIM.
- **Kim, C., Ryu, J., & Kim, K.** (October, 2019). *Development of the virtual reality simulation using problem avatar student for pre-service teachers* [Poster]. 2019 AECT. [\[PDF\]](#)
- **Jeong, M., Kim, C., Ikuji, A., Koga, J., & Umeda, T.** (August, 2018). *Development of scenario for parent counseling simulation* [Poster]. 2018 ICoME. [\[PDF\]](#) **(Young Scholar Award)**

Artificial Intelligence Chatbots (ARAM bot)

- Designed and developed two AI chatbots using Google Dialogflow (IRB informative, argumentative interaction).
- Brainstormed and classified potential questions (intents) and produced responses.
- Investigated examples of Natural Language Processing and conducted a pilot test for developing an automatic evaluation model for human-chatbot interaction.
- **Kim, K., Jung, Y., Kim, C., Jung, M., & Ryu, J.** (May, 2021). *A study on user clustering technique for adaptive chatbot development* [Poster]. Spring Conference of KSET. **(Best Paper Award)**
- **Jung, M., Kim, C., Kim, K., & Jung, Y.** (May, 2021). *Development of data collection adaptive chatbot for machine learning application* [Media Showcase]. Spring Conference of the KSET.
- **Kim, K., Kim, C., Cho, M., & Ryu, J.** (August, 2020). *The automatic evaluation model for argumentative interaction in debating chat-bot program* [Round Table]. 2020 ICoME. [\[PDF\]](#) **(Young Scholar Award)**
- **Kim, K., Kim, C., Cho, M., & Ryu, J.** (May, 2020). *The automatic evaluation model based on artificial intelligence for argumentative interaction in debating chat-bot* [Poster]. Spring Conference of KSET.

- Kim, K., **Kim, C.**, Cho, M., & Ryu, J. (March, 2020). *The automatic evaluation model for interaction analysis with artificial intelligence-based debating chat-bot program* [Concurrent]. 2020 Spring Conference of KAEIM. [\[Video\]](#)

Stealth Assessment in Augmented Reality Simulation (Career Interest Test Game) [\[Link\]](#)

- Designed and developed AR simulation of career interest test for elementary students via stealth assessment.
- Developed models (competency model, evidence model, action model) based on the evidence-centered design.
- Collected user log data (demographic data, selection behavior, etc.) in real-time through Firebase.
- **Kim, C.**, & Ryu, J. (November, 2019). *A study on the application of stealth assessment to career interest test - establishment of evidence-centered-design model applied with bayesian network* [Poster]. Autumn Conference of KSET & KAEIM. [\[PDF\]](#)
- Xu, C., Ryu, J., **Kim, C.**, & Ju, E. (July, 2019). *Implementation of stealth assessment to career interest test for elementary student* [Concurrent]. 2019 ICoME. [\[PDF\]](#) (Young Scholar Award)
- Xu, C., **Kim, C.**, Min, C., & Ju, E. (June, 2019). *Development of AR content for career interest test using stealth assessment* [Media Showcase]. 2019 Spring Conference of the KAEIM. [\[PDF\]](#) (1st Prize at Media Festival)

Motion Capture-based Virtual Reality Simulation for Rehabilitation (MARVEL)

- Utilized Perception Neuron Motion Capture hardware for capturing user's whole-body movement in real time.
- Implemented real-time data linkage between VR (user) and mobile application (human rater) through Firebase.
- Developed real-time storage/feedback of user performance record through Firebase.
- **Kim, C.**, Lim, S., Kim, D., Lee, Y., & Jang, S. (November, 2019). *Motion-based virtual rehabilitation program for post-stroke disorders* [Media Showcase]. 2019 Autumn Conference of KSET & KAEIM. [\[PDF\]](#)

Virtual Reality Simulations for Medical and Dental Students (SimSEAH [\[Link\]](#) & SimDANVI [\[Link\]](#))

- Developed VR simulations for training communication and clinical skills of medical and dental students.
- Conducted collaborative research projects with medical school and dental college.
- Assessed and modified simulations by evaluating learners' experiences through a mixed methods approach.
- Ryu, J., Kim, K., & **Kim, C.** (October, 2019). *Effects of auditory information and task difficulty on user's presence in haptic-based simulation* [Concurrent]. 2019 AECT. [\[PDF\]](#)
- Yoon, H., & **Kim, C.** (November, 2018). *A study on the usability evaluation of ophthalmological strabismus simulation based on virtual reality* [Poster]. 2018 Autumn Conference of KSET & KAEIM.
- **Kim, C.**, Min, C., & Kim, D. (October, 2017). *Ophthalmology career experience simulation using virtual reality technique: Motion-based strabismus measurement program* [Media Showcase]. 2017 Autumn Conference of KSET & KAEIM.

PROFESSIONAL & VOLUNTEER ACTIVITIES

Simulation System for Educating Cross-Eye based on Virtual Reality, Korean Patent No. 10-2406472 Jun. 2022

- Co-developed and patented a VR-based simulation for training ophthalmologists. [\[PDF\]](#)

Visiting Scholar, University of South Florida

Jan. 2020 – Mar. 2020

- Participated in collaborative research led by Dr. Sanghoon Park.
- Designed and developed AR-based training simulation for preservice teachers (SimTEACHER Mobile).
- Submitted book chapter to the AECT Summer Research Symposium.
- Park, S., Ryu, J., Kim, K., & **Kim, C.** (2021). Mobile simulation for effective classroom management skill training for preservice teachers. In B. Hokanson, M. Exter, A. Grincewicz, M. Schmidt & A.A. Tawfik (Eds.). *Learning: Design, Engagement and Definition* (pp. 223-238). Springer, Cham. https://doi.org/10.1007/978-3-030-85078-4_18 [\[PDF\]](#)

Professional Services, KAERA, KELS and EPSA

- **Webmaster**, Korean-American Educational Research Association (KAERA) *Nov. 2024 – present*
- **Board Member**, Korean Edutech/Learning Sciences Researcher Network (KELS) *Oct. 2022 – present*
- **Educational Psychology Student Association**, University of Wisconsin-Madison *Sep. 2024 – Dec. 2024*

Invited Peer Reviewer Services, Journals and Conferences

- **Computers & Education: X Reality**, Journal Reviewer — 1 manuscript *2025-2026*
- **ISLS 2026**, Conference Reviewer — 3 papers *2025*
- **AERA 2026**, Conference Reviewer — 11 papers *2025*
- **ISLS 2025**, Conference Reviewer — 2 papers *2024*

- **KAERA 2025**, Conference Reviewer — 3 papers 2024

Conference Volunteer Services, KAEIM, KSET, AECT, and ICoME

- Volunteered as a moderator at 2025 iLRN. *Jun. 2025*
- Volunteered as a moderator at 2022 AECT. *Oct. 2022*
- Volunteered as a technical support team member at 2021 Autumn Conference of KSET & KAEIM. *Nov. 2021*
- Volunteered as a moderator & technical support team member at 2020 AECT. *Nov. 2020*
- Volunteered as a technical support team member at 2019 AECT. *Oct. 2019*
- Volunteered as a technical support team member at 2018 ICoME. *Aug. 2018*
- Volunteered as a technical support team member at 2018 Spring Conference of KAEIM. *Aug. 2018*

Teaching Volunteer Services

- Served as a **Badger Volunteer**, providing support to over thirty K-5 students at Lussier Community Education Center. *Jan. 2024 – May 2024*
- Volunteered as a student teacher at five educational institutions for underserved students. *Jul. 2016 – Mar. 2019*

PROFICIENCIES & SKILLS

Programming & Tools

- Advanced: SPSS, Wix, WordPress
- Proficient: C Sharp, Unity 3D, Daz 3D, iClone, Vuforia, Firebase, MAXQDA, ELAN
- Intermediate: SQLite, Adobe Illustrator and Photoshop, Adobe Premiere Pro
- Novice: SketchUp

Hardware

- Advanced: Oculus VR
- Proficient: iMotions AFFDEX Module, HoloLens AR
- Intermediate: iMotions Eye Tracking Module, LeapMotion, Perception Neuron Motion Capture Module